# Bat-Orgil Erdenebat

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## Education

**University of Mississippi** — Expected May, 2027

Bachelor of Science in Computer Science – Data Science Emphasis; Engineering Math Minor — **GPA: 3.78**

**Awards:** Freeman Grant - $9,000, Chancellor's Honor Roll Fall 2024, Spring 2025

**Certifications:** Machine Learning Foundations e-Certificate (Cornell University) May - August 2026, Career Essentials in Generative AI (Microsoft) June 2024

## Technical Skills

**Languages:** Python, R, Java, SQL, C++, Rust, JavaScript, TypeScript, C, LaTeX

**Libraries/Frameworks:** PyTorch/Keras, TensorFlow, Hugging Face Transformers, Scikit-learn, OpenCV, Pandas, NumPy, MediaPipe, Flask, Tailwind CSS, Matplotlib, Seaborn

**Tools/Platforms:** Git, GitHub, Neo4j, Google Cloud, MongoDB, MySQL, Docker, Figma, GitLab, AWS

## Professional Experience

**Break Through Tech** — New York, NY

**AI Studio Fellow** — May 2026 – Present

- Selected from **4,000+** applicants nationwide for a one-year AI fellowship focused on machine learning, data science, mentorship, and career readiness.
- Completing a 9-week ML Foundations curriculum using industry-relevant tools to analyze data, build pipelines, and train and validate ML models.
- Preparing for AI Studio challenge projects with industry advisors to deliver real-world ML solutions and portfolio-ready work.

**PotatoNet Co., Ltd.** — Seoul, South Korea

**Machine Learning Engineering Intern** — June 2026 - Present

- Extracted a CNN-LSTM-Attention fault predictor (**94.7%** accuracy) from a legacy Django monolith into a clean FastAPI service on a 16-container microservices platform, and migrated an internal LSTM service to an Nginx → Gunicorn → Django + TLS stack with zero-downtime, reusable deployments.
- Built and shipped a production graph-ETL pipeline that backfilled 6 months of threat data into a live Neo4j store (**19.4K** malicious-JS nodes, 13.7K domains, 3.9K IPs) via idempotent, rollback-tagged loads automated as a nightly self-healing cron job, and deduplicated 267K scripts with SimHash LSH into a leak-free 12.4K-example training set.
- Engineered a 7-signal attacker-profiling engine that clusters malicious JavaScript into named "families" (SimHash AST similarity, perceptual screenshot hashing, and shared TLS-cert/IP/ASN/registrant infrastructure) via GDS Leiden community detection, then diagnosed and fixed a 52K-script over-merge caused by benign library bridges — cutting max family size to **476** across 334 clean families.

**University of Mississippi – Dept. of Computer & Information Science** — Oxford, MS

**Teaching Assistant & Computer Science Tutor** — August 2025 – Present

- Led hands-on labs for **200+** STEM students across four courses (Data Structures and Algorithms), lifting Python, Java, and OOP proficiency by coaching students through debugging, optimization, and algorithmic reasoning.
- Raised academic performance for **50+** mentees by delivering individualized tutoring in Python, Java, and Data Structures & Algorithms, reinforcing programming fundamentals and independent problem-solving.

## Projects

**Census Income Classification** — github.com/BatOrgil7/census-income-ml-pipeline

- Developed an end-to-end binary classification pipeline to predict an individual's annual income by performing exploratory data analysis, data preprocessing, feature engineering, and training traditional machine learning models on the U.S. Census Income dataset.
- Evaluated and optimizing model performance by comparing supervised learning algorithms using accuracy, precision, recall, F1-score, and ROC-AUC, while applying hyperparameter tuning to improve generalization.
- Implemented a neural network model to solve the same classification task and comparing its predictive performance, robustness, and fairness against traditional machine learning approaches.

**Streaming Evaluation Protocol for CSLR** — github.com/BatOrgil7/signstream-eval

- Designed a standardized evaluation protocol for streaming sign language recognition, adapting latency/stability metrics from streaming ASR and simultaneous MT to fill a gap left by offline-only benchmarks.
- Constructed a fault-tolerant landmark extraction pipeline (multiprocessing, content-addressed caching, failure-manifest tracking) processing PHOENIX-2014T across video and frame-folder formats.